Meet 0

October 19, 2026

Algebra

You have 20 minutes to complete 6 problems. There is no penalty for guessing. Answers are only valid if they are written in the below table.

First and last name	

p	Answer	p	Answer
1		4	
2		5	
3		6	

1. Alice and Bob are talking to each other. Bob says, "the square root of the number I'm thinking of is a positive integer strictly less than 5." Alice says, "ok rbo" and leaves. If Bob was telling the truth, what is the sum of all possible values of the number he is thinking of?

2. Two runners start running on a 100 meter track. Runner A runs at a rate of 10 meters per second and runner B runs at a rate of 9 meters per second. Runner A starts 100 meters from the finish, and runner B starts x meters ahead of runner A. If runners A and B reach the end at the same time, compute x.

3. Steven's clock loses 6 minutes every hour. He sets it to the correct time at exactly noon. Later that day, the clock shows exactly 6:00 PM. How many minutes after 6:00 PM is the actual time?

4. Given that the sum of the interior angles of an n-gon, $3n$-gon, $5n$-gon, $\ldots$, $89n$-gon equals the sum of the interior angles of a $(2026n - 101)$-gon, find n.

5. Suppose the graphs

$$y = |2x|$$
$$y = -|3x - 90| + c$$

   intersect at exactly one point for some value of c. Compute c.

6. Let a and b be integers. Suppose that for some positive integer x, $x^{2026} + ax + 1 = 5$ and $x^{2026} + bx + 1 = 17$. Find the sum of all possible integer values of $b - a$.